

The Grand Challenge of Quantum Applications

Paper Discussion

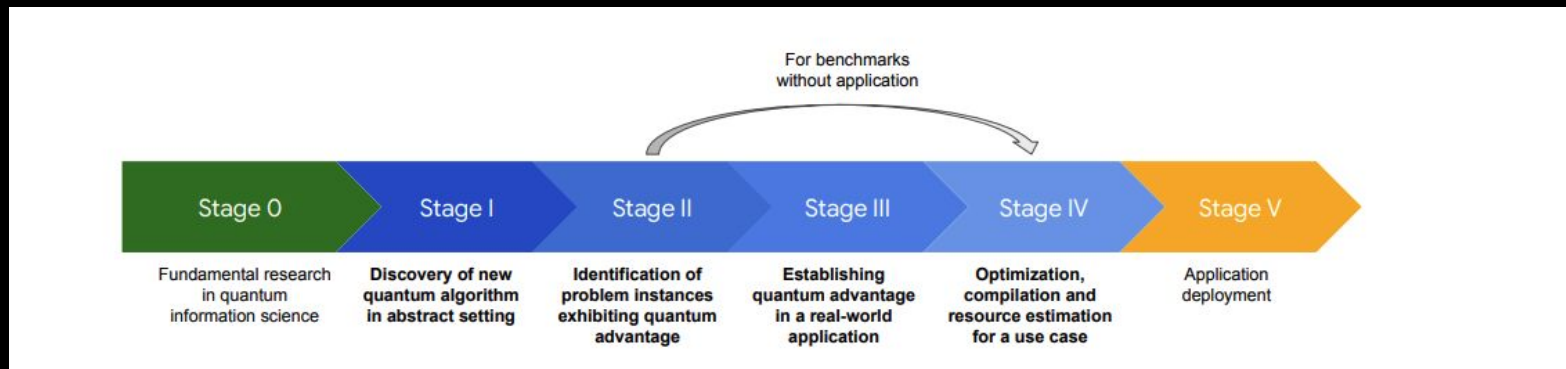
Based on Babbush *et al.* (2025)

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Paper's Motivation

What kind of work needs to happen for useful quantum applications to actually materialize?



The biggest bottleneck is not hardware, but finding real applications (Stage II and Stage III).



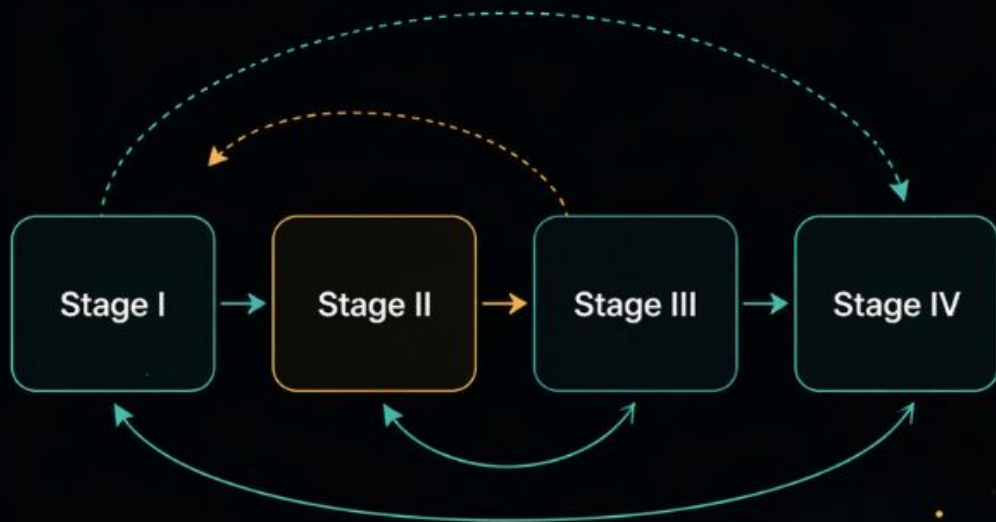
Critiques

The paper suggests stages are descriptive, not sequential.

Steps are often skipped or blend in together.

However that can also be viewed as a limitation of this framework:

Is it a roadmap or classification?
Does it explain how applications actually move forward?



Discovery is messier than the diagram suggests.

Applications



1. Quantum chemistry /
molecular energy estimation

No broad advantage shown



2. Quantum simulation for
materials design

Promising, but case-specific



3. Quantum random number
generation

Closest to deployment, narrow benefit



4. Cryptography /
cryptanalysis

Strongest theoretical case



5. Nuclear reactions /
fusion

Exploratory simulation example

Discussion Questions

1. Are we asking the wrong question if we focus only on computational speed?
2. Should sensing, measurement, or hybrid scientific discovery count as quantum applications.
3. How much evidence should be required before a Stage III claim is convincing?
4. Is the real bottleneck technical, or the lack of cross-domain application search?



Thank you!

